

Executive Summary

Corrected Technical Position

The notebook uses leakage-safe cumulative historical CTR features with an expanding smoothing prior. It separates model training, threshold tuning, and final holdout evaluation by time.

Temporal Split

Training records end before 2017-07-06; threshold tuning uses 2017-07-06; final evaluation uses the untouched 2017-07-07 holdout.

Threshold Search

Threshold tuning now searches the full 0.01 to 0.99 range. XGBoost still selects 0.500, confirming 0.500 is not just the old upper-bound cap.

Final Model

XGBoost - weighted selected by the predefined F1-based rule. ROC-AUC 0.594; PR-AUC 0.082; F1 0.142; recall 0.361; precision 0.088.

Final Training Strategy

The selected architecture and threshold are frozen after temporal validation; the final scoring model is retrained on all available labeled history before predicting the test set.

Implemented Feedback Validation

- ? Fixed subtle prior leakage using an expanding prior inside cumulative training CTR features.
- ? Fixed threshold overfitting by tuning thresholds on a middle date and reporting final metrics on the latest date.
- ? Expanded threshold search from the capped 0.03-0.50 range to 0.01-0.99.
- ? Retrained the selected final architecture on all available labeled history after freezing the threshold.
- ? Kept probability language as click-propensity score because calibration is imperfect.

Model Comparison ? Untouched Final Holdout

	model	threshold	roc_auc	pr_auc	f1	precision	recall	tn	fp	fn	tp
XGBoost - weight		0.5000	0.5935	0.0819	0.1417	0.0881	0.3611	50626	16439	2812	1589
XGBoost - weighted - tune		0.5000	0.5935	0.0819	0.1417	0.0881	0.3611	50626	16439	2812	1589
Logistic Regression - balanced - tune		0.4650	0.5955	0.0817	0.1395	0.0831	0.4329	46051	21014	2496	1905
Logistic Regression - balanced - tune		0.5000	0.5955	0.0817	0.1359	0.0904	0.2740	54928	12137	3195	1206
Random Forest - balanced - tune		0.4650	0.5901	0.0810	0.1344	0.0874	0.2904	53720	13345	3123	1278
Global CTR probability		0.0704	0.5000	0.0616	0.1160	0.0616	1.00	0	67065	0	4401
Random Forest - balanced - tune		0.5000	0.5901	0.0810	0.1049	0.0990	0.1116	62594	4471	3910	491
Always non-click baseline		0.5000	0.5000	0.0616	0.0000	0.0000	0.0000	67065	0	4401	0

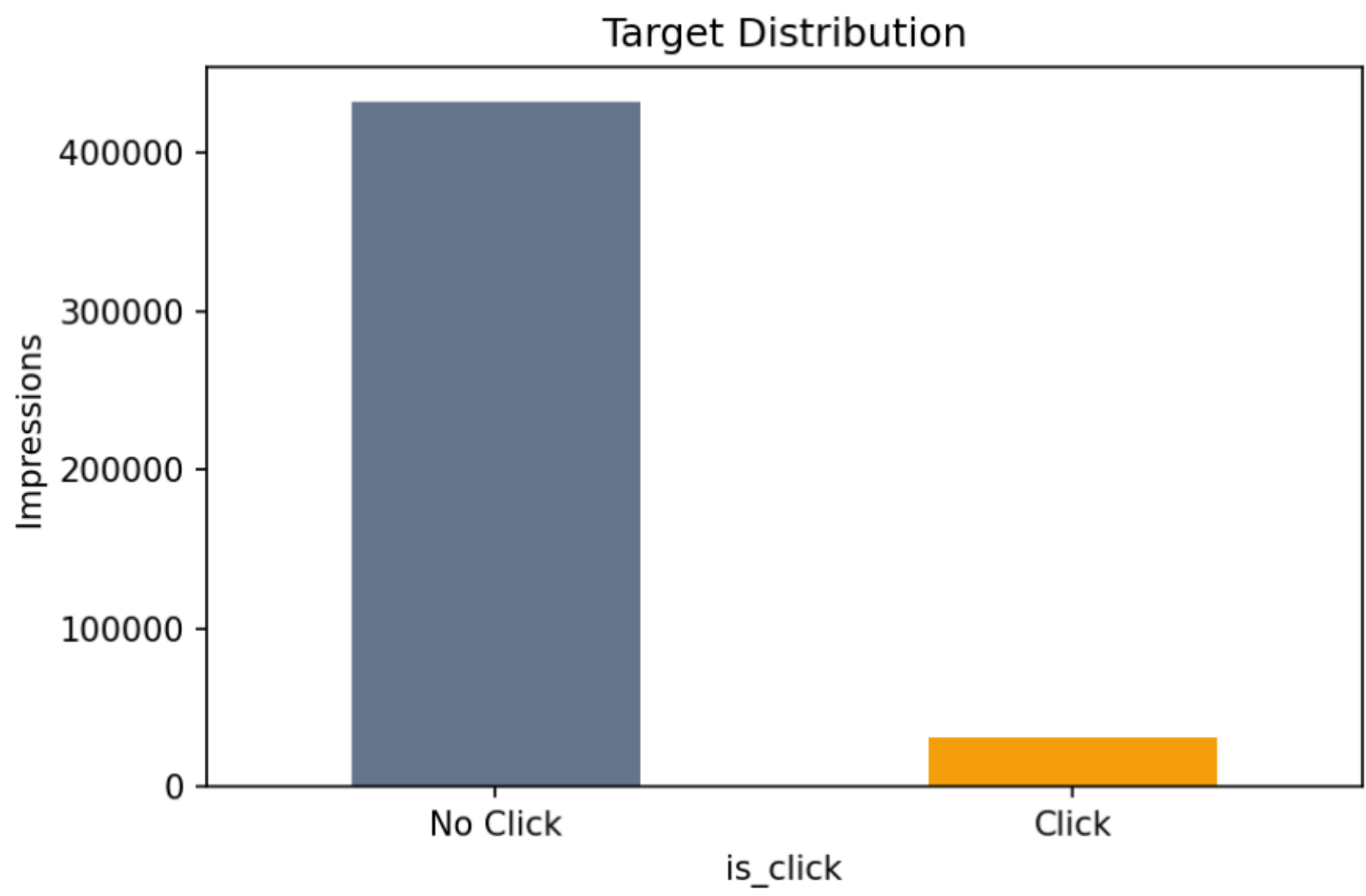
Thresholds are selected on the middle temporal split using a 0.01-0.99 grid, not on this final holdout.

Feature Ablation ? Incremental Value

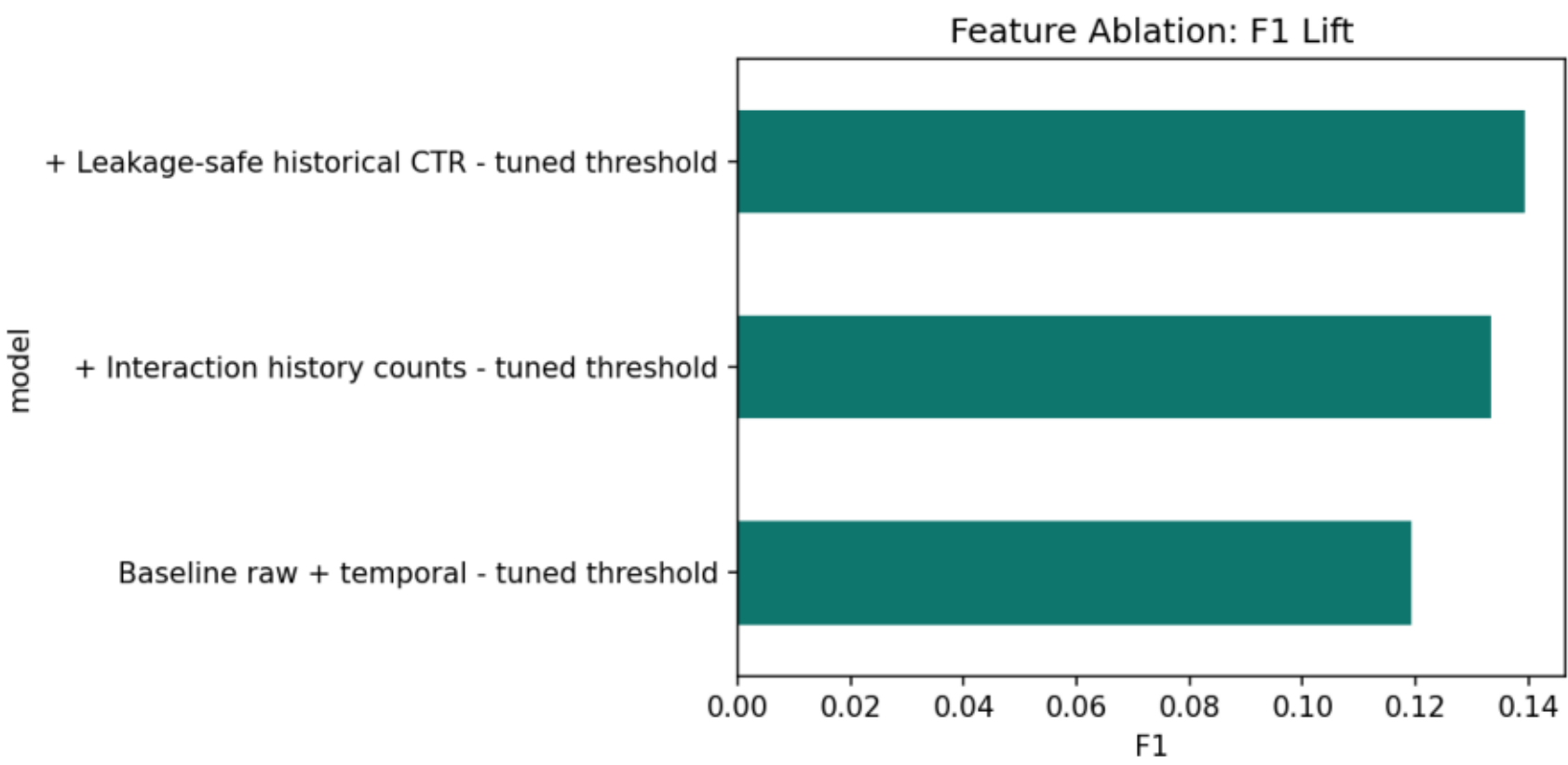
	model	threshold	roc_auc	pr_auc	f1	precision	recall
Baseline	raw + temporal - tuned	0.4800	0.5301	0.0676	0.1193	0.0660	0.6199
+ Interaction	history counts - tune	0.5150	0.5759	0.0762	0.1334	0.0782	0.4549
+ Leakage-safe	historical CTR - tune	0.4650	0.5955	0.0817	0.1395	0.0831	0.4329

Interaction history and leakage-safe CTR features add predictive value over raw + temporal features.

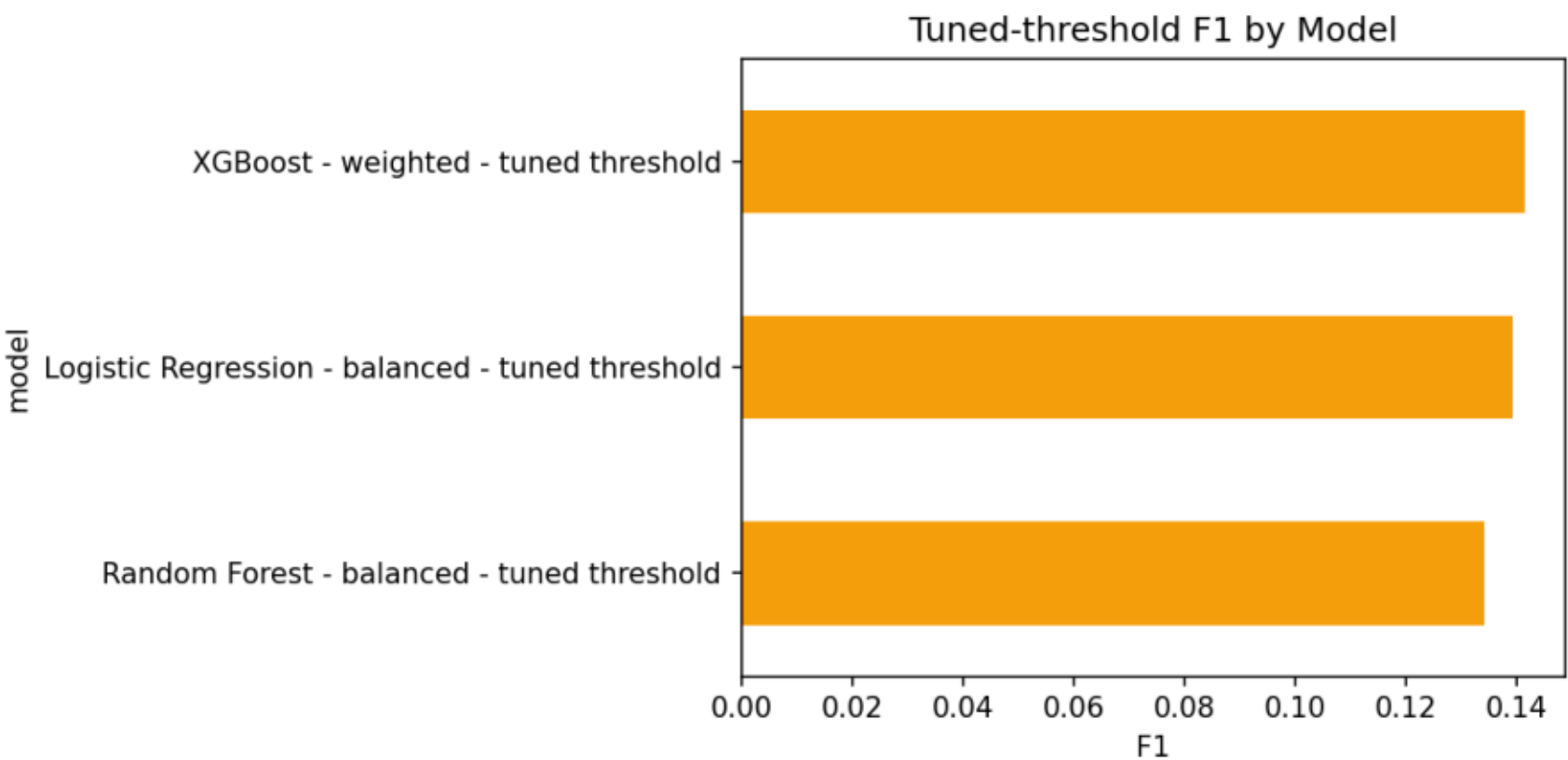
Target Distribution



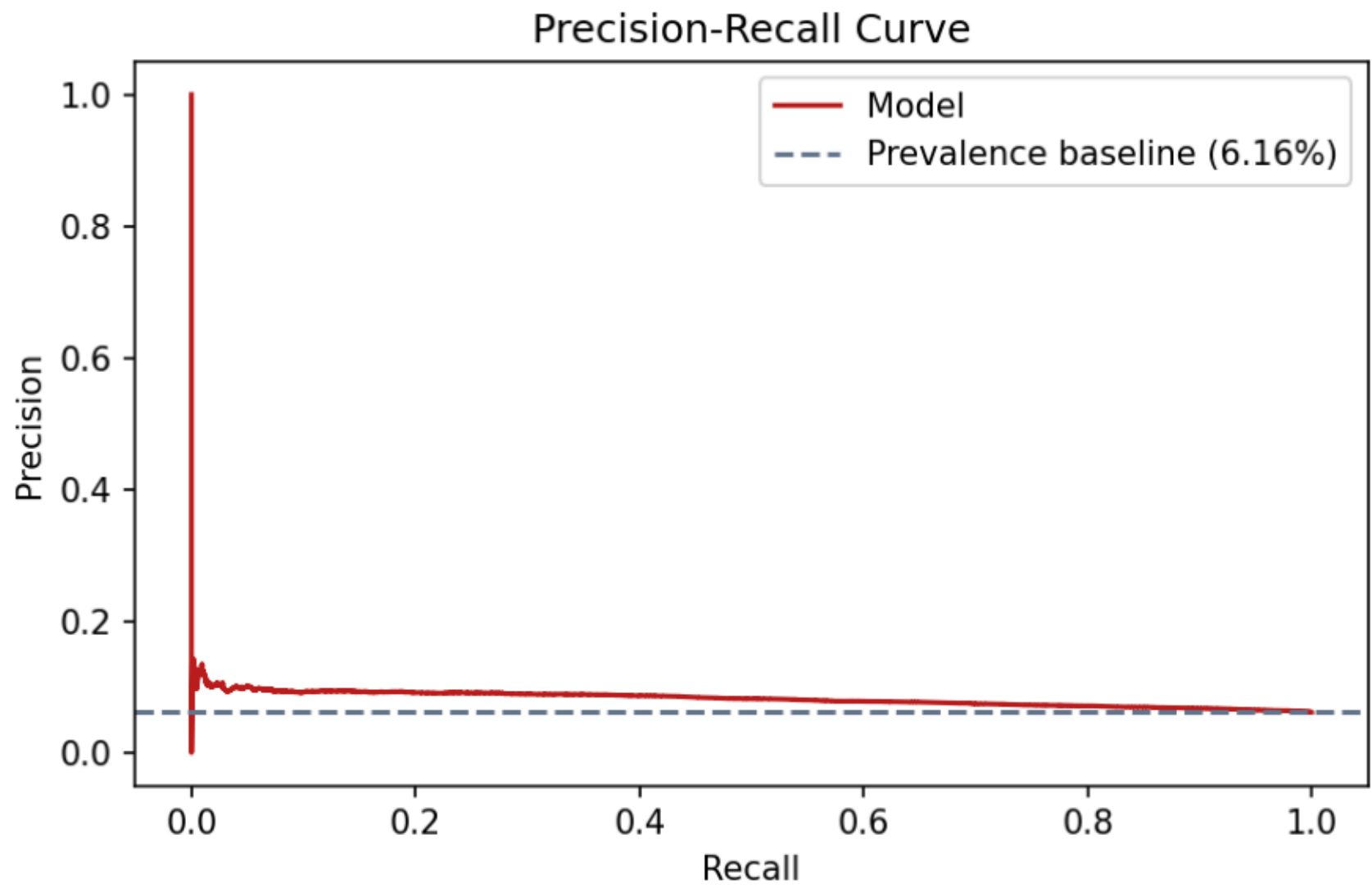
Feature Ablation Chart



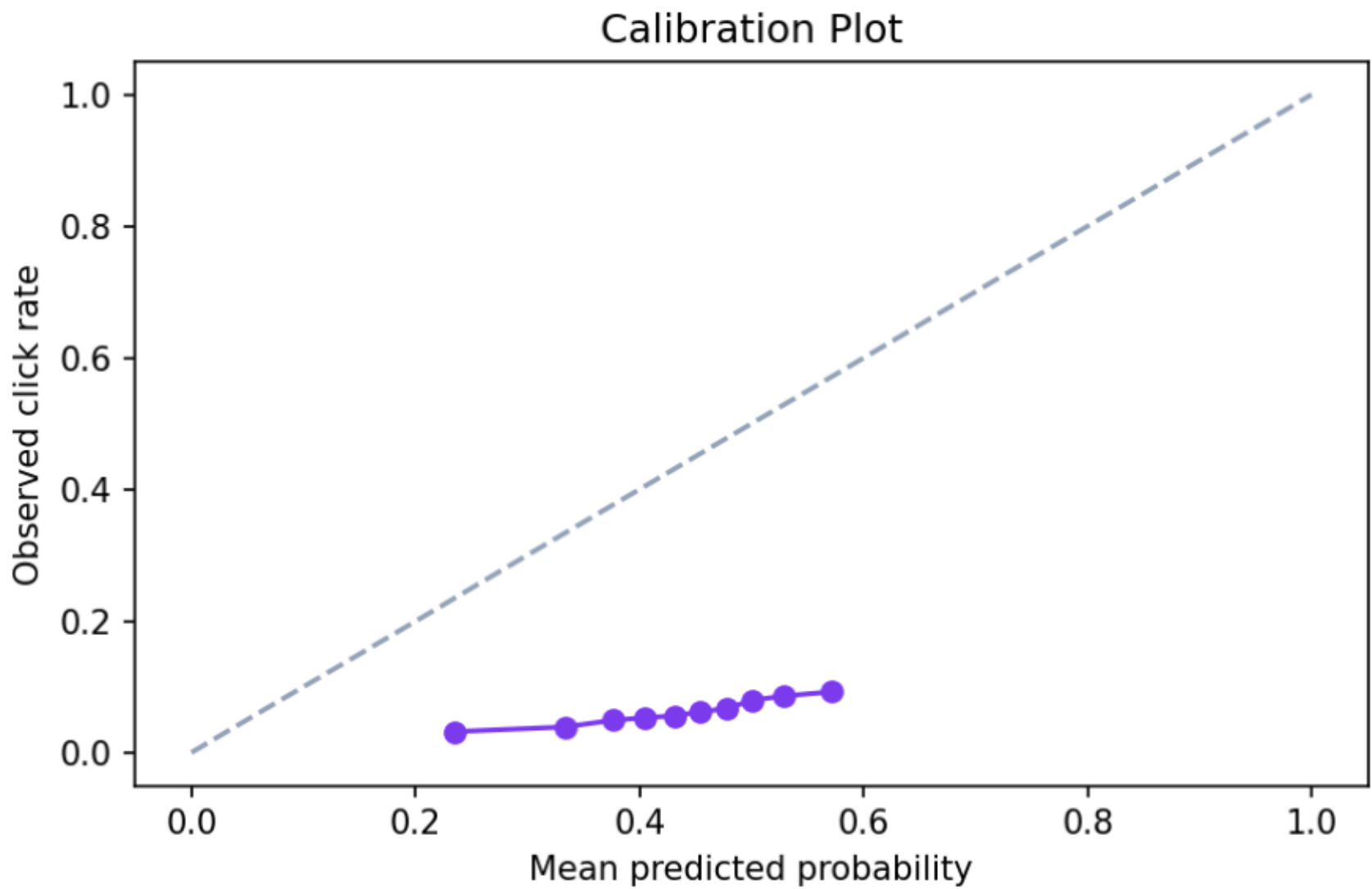
Model F1 Comparison



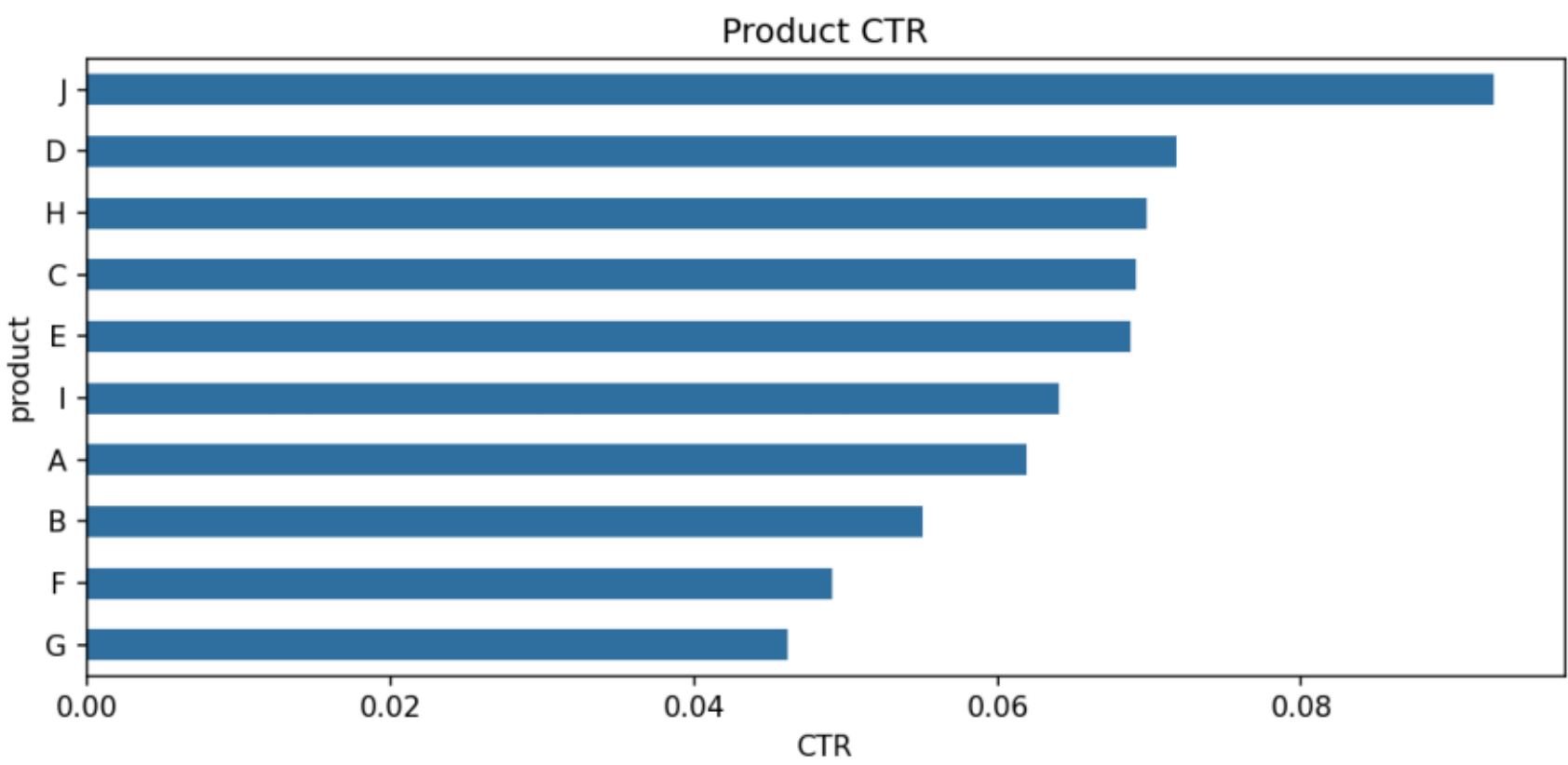
Precision-Recall Curve with Prevalence Baseline



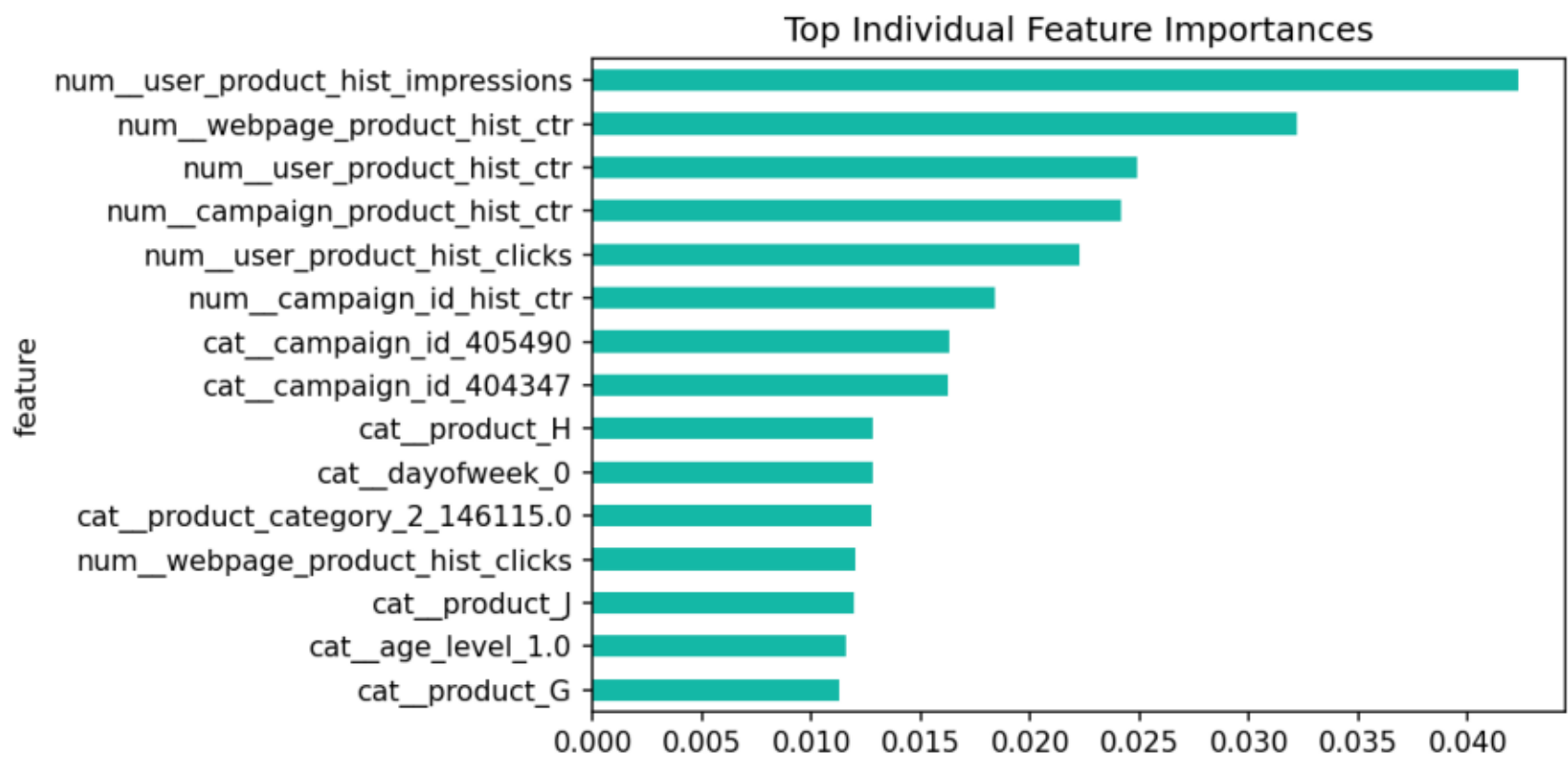
Calibration Plot



Product CTR / Inventory Planning Signal



Top Individual Feature Importances



Top Product CTR Planning Signal

product	clicks	impressions	ctr	expected_clicks_per_100k_impressions
J	899	9698	0.0927	9,269.95
D	2949	41064	0.0718	7,181.47
H	7654	109574	0.0699	6,985.23
C	11306	163501	0.0691	6,914.94
E	1474	21452	0.0687	6,871.15
I	4079	63711	0.0640	6,402.35
A	953	15391	0.0619	6,191.93
B	1238	22479	0.0551	5,507.36
F	344	7007	0.0491	4,909.38
G	435	9414	0.0462	4,620.78

This is an inventory-planning signal, not a formal demand forecast.

High-Propensity User Profiles with Wilson Intervals

gender	age_level	city_development_index	clicks	impressions	ctr	ctr_ci_low	ctr_ci_high
Male	5.00	4.00	255	2945	0.0866	0.0770	0.0973
Female	5.00	4.00	100	1181	0.0847	0.0701	0.1019
Male	1.00	2.00	605	7922	0.0764	0.0707	0.0824
Female	5.00	2.00	208	2752	0.0756	0.0663	0.0861
Male	2.00	2.00	3736	50409	0.0741	0.0719	0.0764
Male	1.00	3.00	607	8291	0.0732	0.0678	0.0790
Male	1.00	4.00	306	4252	0.0720	0.0646	0.0801
Male	5.00	3.00	349	4909	0.0711	0.0642	0.0786
Male	2.00	3.00	1927	27671	0.0696	0.0667	0.0727
Male	5.00	2.00	493	7112	0.0693	0.0636	0.0755

ML Insight to Business Action

Model insight	Business action
High product CTR	Increase exposure and reserve more inventory
High webpage CTR	Prioritize placement and monitor page quality
High user-product historical CTR	Personalize ad selection and test targeted bid uplift
Weekend uplift	Test weekend bid modifier before permanent adoption
High predicted probability	Prioritize impression under budget constraints
Low CTR product	Refresh creative or reduce allocation
Model drift	Retrain and recalibrate thresholds

Recommendations are framed as tests or ranking actions because model discrimination/calibration are moderate.

Business Questions ? Updated Answers

? Weekend CTR is 7.33% vs weekday CTR 6.65%; uplift is 0.68 pp / 10.2% relative; p-value 4.27e-12.
Test ROI before permanent bid uplift.

? Highest CTR products are J, D, H; lowest CTR products are B, F, G.

? Ablation confirms interaction history and historical CTR features improve model lift over the raw + temporal baseline.

? Top individual features show user-product and contextual historical behavior are central click-propensity signals.

? SMOTE was not experimentally evaluated in the final pipeline because categorical-aware SMOTENC is required; class weighting was selected as the practical imbalance approach.

? Product CTR should be treated as expected clicks per 100K impressions for planning, not a true demand forecast.

? User-profile bid actions require confidence intervals, minimum volume, fairness/privacy review, and live drift monitoring.

Limitations and Next Steps

- ? The final model improves over baselines but remains below aspirational ROC-AUC/F1/precision targets.
- ? Apply Platt scaling or isotonic regression before treating scores as literal click probabilities.
- ? The final scoring model now retrains on all labeled history; production can scale this further with distributed training.
- ? Add richer behavioral, creative, bid, placement quality, and recency features if available.
- ? Run controlled A/B tests before converting offline click-propensity gains into bidding rules.